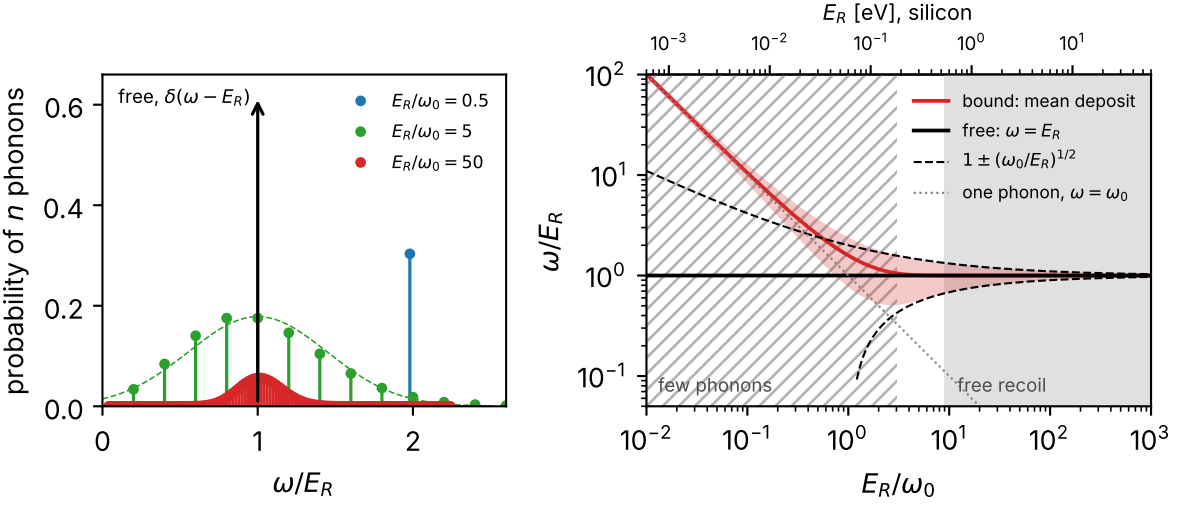
probability of n phonons
free, $\delta(\omega - E_R)$
$E_R/\omega_0 = 0.5$
$E_R/\omega_0 = 5$
$E_R/\omega_0 = 50$
0.0
0.2
0.4
0.6
0
1
2
ω/E_R
E_R [eV], silicon
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
ω/E_R
10^{2}
10^{1}
10^{0}
10^{-1}
bound: mean deposit
free: $\omega = E_R$
$1 \pm (\omega_0/E_R)^{1/2}$
one phonon, $\omega = \omega_0$
few phonons
free recoil
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
E_R/ω_0